

Homework 4
MSBA 400: Statistical Foundations for Data Analytics
Professor Rossi

Question: Prediction of Catalogue Orders

The dataset `cat_buy.rda` contains data on the response of customers to the mailing of spring catalogues. The variable `buytabw` is 1 if there is an order from this spring catalogue and 0 if not. This is the dependent or response variable (literally was there a “response” to or order from the direct mailing).

This spring catalogue was called a “tabloid” in the industry. The catalogue featured women’s clothing and shoes. The independent variables represent information gathered from the internal `house file` of the past order activity of these 20,617 customers who received this catalogue.

In direct marketing, the predictor variables are typically of the “RFM” type: 1. Recency 2. Frequency and 3. Monetary value. This data set has both information on the volume of past orders as well as the recency of these orders.

The variables are:

- * `tabordrs` (total orders from past tabloids)
- * `divsords` (total orders of shoes in past)
- * `divwords` (total orders of women’s clothes in past)
- * `spgtabord` (total orders from past spring cats)
- * `moslsdvs` (mos since last shoe order)
- * `moslsdvw` (mos since last women’s clothes order)
- * `moslstab` (mos since last tabloid order)
- * `orders` (total orders)

part A

Use the R `sample` command to randomly sample 1/2 of the data. The `sample` command will sample randomly from a list of numbers, e.g. `sample(1:10,size=5)` will select 5 numbers at random from the numbers 1,2,3,4,5,6,7,8,9,10.

Use `sample` to select row numbers and then use these row numbers to divide your data into two parts. One part for estimation and one part for validation.

Hint: see code below (modify)

```
ind.est=sample(???,???)
est_sample = cat_buy[ind.est,]
holdout_sample = cat_buy[-ind.est,]
```

part B

Fit a logistic regression model using the estimation sample produced in part A. Eliminate insignificant variables.

Discuss your final specification, do the signs of the coefficients make sense to you?

Should you worry about multi-collinearity in this dataset?

part C

Use the best-fit from part B to predict using the holdout sample.

Plot boxplots of the fitted probabilities for each value of `buytabw` for the holdout sample (see code snippets from Chapter 7 for an example)

Compute a “lift” table as done in Chapter 7 code snippets.